Joyce (Jiusi) Wang

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SUMMARY

EE student at Georgia Tech (4.0 GPA) focused on hardware engineering: ASIC physical design, design verification, mixed-signal PCB design and hardware test, with hands-on RTL-to-GDS work in Cadence Innovus and UVM testbenches in SystemVerilog.

EDUCATION

Georgia Institute of Technology Atlanta, GA

B.S. Electrical Engineering | *GPA: 4.0/4.0* | *Threads: Signal & Information Processing, Circuit Technology* *Expected May 2028*

Relevant Coursework: Digital System Design (ECE 2020), Programming for Hardware/Software Systems (ECE 2035), Circuit Analysis (ECE 2040)

TECHNICAL SKILLS

ASIC Design & Verification: SystemVerilog, RTL Design, Universal Verification Methodology (UVM), SystemVerilog Assertions (SVA), Functional Coverage, Cadence SimVision, Cadence IMC

Physical Design: Cadence Innovus, RTL-to-GDS, Floorplanning, Place and Route, Clock Tree Synthesis (CTS), Timing Closure, DRC Debug

Hardware & PCB: Altium Designer, Schematic Capture, PCB Layout, Flexible PCB, LTspice, LDO Regulators, ADCs, Instrumentation Amplifiers, PCB Assembly, BGA Reflow & Hot-Air Rework

Programming: Python, Tcl, MATLAB, Simulink, C/C++, Bash, Java | **Languages:** English (Native), Mandarin Chinese (Native)

EXPERIENCE

ASIC Physical Design Engineer – Silicon Jackets Jan 2026 – Present | Atlanta, GA

- Ran RTL-to-GDS place-and-route in Cadence Innovus on a 64-bit ASIC, resolving 1,000+ DRC violations to zero via SRAM LEF obstruction analysis, 15-unit placement halos, MET1–MET3 routing blockages, and metal layer expansion.
- Closed setup timing from −2.709 ns to +0.01 ns WNS and hold timing from −0.032 ns to 0 ns; wrote Python report parsers and Tcl procedures to automate timing analysis.
- Starting physical design of Vortex, an open-source RISC-V GPU, in Cadence Innovus (Oct 2026).

Research Assistant – GAMMA Research Group, Georgia Tech Dec 2025 – Present | Atlanta, GA

- Built a MATLAB surrogate-optimization pipeline for OTA circuit sizing, cutting circuit evaluations 5× (500 vs. 2,511) and runtime about 8× (3 min vs. 25 min) with the same optimal design; flagged a figure-of-merit formula error.

Student Researcher – Mechatronics & Motivation Lab, Georgia Tech Jan 2026 – May 2026 | Atlanta, GA

- Co-designed with the EMG subteam lead a 2-layer flexible PCB in Altium, integrating separate analog and digital 3.0 V LDO supplies, a 24-bit ADS1298 biopotential ADC and a BLE module for real-time EMG acquisition.
- Hand-assembled and reworked 3 prototype PCBs using BGA reflow, flux and hot-air soldering under a stereo microscope; traced a clock-select pin strap error to the datasheet, fixed it with a jumper wire, and wrote an assembly guide.

Technical Lead – NASA Student Launch Jun 2023 – Jun 2024 | Minnesota (Hybrid)

- Led payload design, integration and flight-readiness testing for a dual-deployment high-power rocket, working closely with NASA engineers; achieved a 1-mile altitude launch, qualifying from 1,000+ competing teams.

R&D Intern Engineer – Shanghai ZhiCheng Software Co., Ltd. Summer 2024 | Shanghai, China

- Developed embedded control software processing 5 real-time vehicle inputs (chassis height, speed, brake, throttle, steering) to actuate an air pump and hold chassis height within ±5 mm across 4 driving conditions; built a Bluetooth diagnostics app (software copyright).

PROJECTS

RISC-V (RV32I) Single-Core CPU | SystemVerilog, RTL-to-GDS Sep 2026 – Present

- Designing and verifying a 3-cycle RV32I-subset CPU (fetch, decode, execute) in SystemVerilog supporting addi, add, sub, lw, sw, beq, sll, srl and ebreak; integrating SRAM and memory-controller interfaces and building the RTL-to-GDS flow.

64-Bit Calculator ASIC | Hierarchical RTL-to-GDS Design Jan 2026 – Mar 2026

- Built a parameterized UVM testbench reaching 99.82% functional coverage and a 100% SVA pass rate across 7 assertions.
- Designed the SystemVerilog architecture with a 7-state FSM controller, parameterized ripple-carry adder hierarchy and dual SRAM macros.

Wheatstone Bridge Sensor Circuit | LTspice, Breadboard Test Fall 2026

- Added an AD8221 instrumentation amplifier (gain 100) to a Wheatstone bridge in LTspice, raising sensitivity from 1.25 mV/Ω to 125 mV/Ω; built and bench-tested the passive bridge.

LEADERSHIP & HONORS

RHA Meetings Coordinator | GT Registered Peer Tutor | Founder, bilingual tutoring business (5+ years tutoring)

IEEE Eta Kappa Nu Member | Salutatorian | Presidential Scholar Nominee | Genius Olympiad International Finalist | Most Outstanding STEM Exhibit (Yale) | Naval Science Award (ONR) | NOAA Award | State Science Fair 2nd Place